K-Means Clustering Using Plasma Colors

Full Python Program

```
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.cluster import KMeans

x, y = make_blobs(n_samples=300, centers=4, random_state=42)

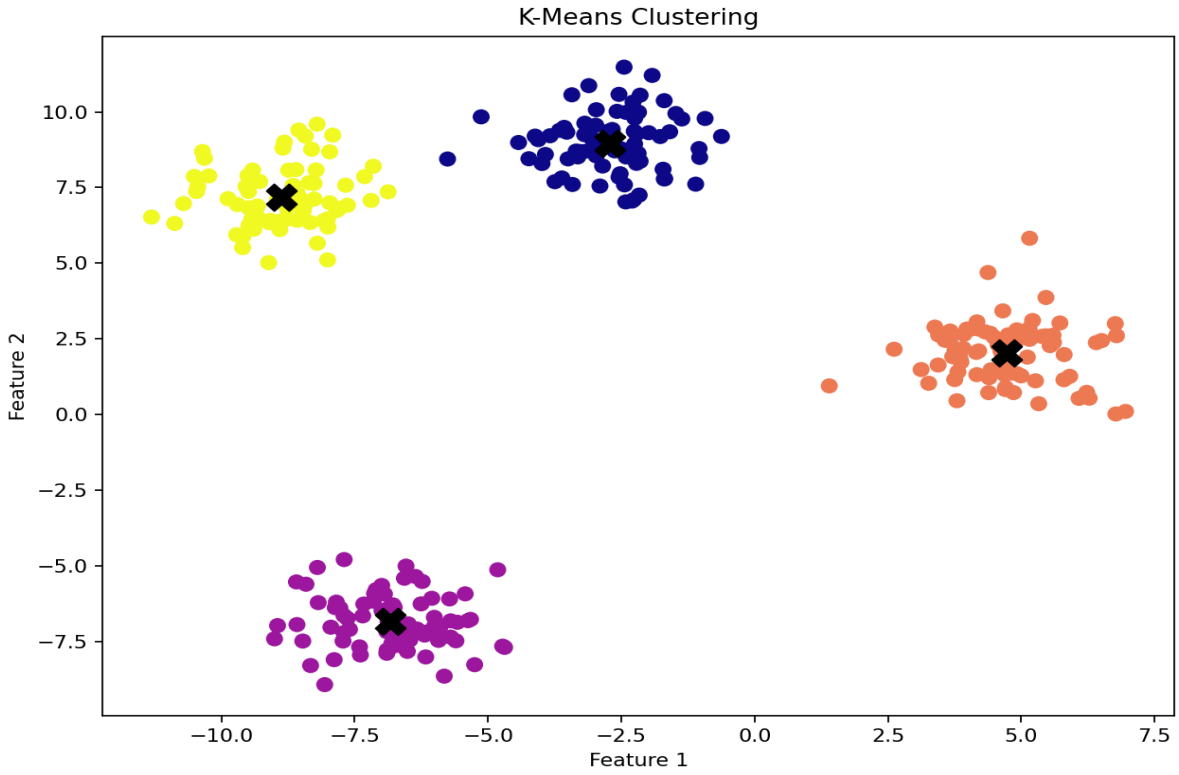
model = KMeans(n_clusters=4, random_state=42, n_init=10)
model.fit(x)

plt.scatter(x[:, 0], x[:, 1],
            c=model.labels_, cmap='plasma', s=50)

plt.scatter(model.cluster_centers_[:, 0],
            model.cluster_centers_[:, 1],
            color='black', marker='X', s=200)

plt.title("K-Means Clustering")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```

Full Output



Result: The K-means algorithm groups 300 data points into four clusters. Different plasma colors show the clusters, and black X marks show their centers.

Install the required libraries if needed: `pip install matplotlib scikit-learn`